MUHAMMAD AHMED IMTIAZ

AI/ML Engineer

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Summary

AI/ML Engineer building production biometric identity verification systems. Core engineer on a multi-tenant KYC platform combining face matching, liveness detection, OCR and fingerprint recognition as Python/FastAPI microservices, with Go backends, Kafka-based event pipelines, PostgreSQL and Redis. Experienced across the full lifecycle: model integration and evaluation (FAR/FRR/EER, AUC/F1), image preprocessing, secure API design (authentication, RBAC, API keys, signed webhooks), and Docker-based deployment. Background in PyTorch, Hugging Face Transformers, scikit-learn and RAG-based retrieval from earlier projects in voice AI agents, recommendations and smart farming.

Technical Skills

- **Primary / Confident:** Python · PyTorch · scikit-learn · OpenCV · FastAPI · Microservices · Deep Learning · CNNs (EfficientNet) · Transformers (BERT) · Biometrics (face, liveness, fingerprint: DeepPrint) · Image enhancement (CLAHE, ridge analysis) · Fine-tuning & LoRA · TensorFlow · Hugging Face · FAISS · RAG · XGBoost · Rasa · Embeddings · Model evaluation (AUC / F1, FAR / FRR / EER) · InsightFace · PaddleOCR · Git / GitHub / Bitbucket · REST API design · OpenAPI / Swagger · Authentication & RBAC (Ed25519 tokens, API keys) · Webhooks (HMAC signing) · API gateway (Apache APISIX) · PostgreSQL · Redis · Apache Kafka · Docker · Go (Golang) · SQL · Docker Compose · Linux / Bash · Claude Code · Cursor · Windsurf
- **Familiar / Working knowledge:** Next.js · n8n · Open-source STT/TTS (whisper.cpp, Vosk, Coqui/Piper) · MySQL · Flask · React · C++ · Java
- **Professional Skills:** Problem-solving · Analytical thinking · Technical communication · Team collaboration · Time management · Adaptability

Experience

AI/ML Engineer, Neuronix Technologies *Sep 2026 – Present*

- Core AI/ML engineer for KYC on a multi-tenant identity verification platform (KYC, KYB, transaction screening) serving external integrators via public REST APIs.
- Designed the platform's security layer: shared Go auth library with Ed25519-signed tokens and Redis-backed sessions (instant revocation), hierarchical RBAC, per-tenant hashed API keys with rate limiting, and API-gateway (APISIX) hardening.
- Built a signed webhook system (HMAC-SHA256, transactional outbox, retries with backoff) and an async verification API with idempotent request IDs, a job queue with leases, and an OpenAPI 3.1 specification.
- Investigated and fixed a fingerprint false-accept issue: traced it to a scanner-trained model (DeepPrint) applied to phone images, unified decision thresholds, and added an image validation gate (blur, ridge-pattern checks) and enhancement pipeline (CLAHE, scale normalisation).
- Led R&D on contactless fingerprint recognition: evaluated models (Ridgeformer, C2CL) and datasets (RidgeBase, ISPFD, HKPolyU, synthetic IMPOSE) for licensing and fine-tuning strategy (partial fine-tuning, LoRA).

AI/ML Intern, Neuronix Technologies *Mar 2026 – Aug 2026*

- Built the core biometric verification services as Python/FastAPI microservices: face matching (InsightFace), liveness detection (ViT), OCR (PaddleOCR) and fingerprint matching (DeepPrint).
- Designed an event-driven pipeline with Apache Kafka: Go data-acquisition service calling AI services in parallel, Go consumer persisting results to PostgreSQL, and a Go/Gin backend serving Next.js admin and client portals.
- Containerized and deployed the full stack with Docker Compose on Linux, resolving production issues in Kafka networking, model memory limits and service orchestration.

AI/ML Intern, Codecelix NASTP *Nov 2025 – Feb 2026*

- Team lead for applied ML prototypes built with open-source tools.
- **Voice Customer Support Agent:** built a low-latency voice-to-voice pipeline (speech $\rightarrow$ text $\rightarrow$ LLM response $\rightarrow$ speech) with open-source STT, LLM and TTS, in a modular design.
- **EchoSee:** developed AI accessibility and safety features: sound-event detection, 360° directional audio awareness, and a real-time lip-reading prototype for captions.

AI Intern, Dam Techhub *July 2025 – August 2025*

- Built a graph-based product recommender on the Amazon co-purchase graph: Node2Vec embeddings and a GraphSAGE link-prediction model (AUC $\approx$ 0.85 on validation/test).
- Mapped internal node IDs to real Amazon ASINs for frontend use.
- Served recommendations through a FastAPI `/recommend` endpoint with JWT authentication and environment-based configuration.

Projects

Agentic RAG-based Smart Farming Advisor *2025*

- Built a multi-model agricultural assistant with a rule-based router dispatching queries to three modules (routing accuracy $\approx$ 0.88).
- Crop recommendation with XGBoost / RandomForest on tabular data, with scaling, encoding and feature selection (test accuracy $\approx$ 0.86).
- Leaf-disease detection with a transfer-learned EfficientNet CNN and data augmentation (test accuracy $\approx$ 0.92).
- RAG pipeline: document chunking, embeddings, FAISS index and a generator fallback for low-recall queries (hit-rate@5 $\approx$ 0.78).

Feedback Flow (Final Year Project) *2024 – 2025*

- Fine-tuned a BERT sentiment-analysis model on Uber reviews and served it to a Flutter app through Flask APIs.
- Built a Rasa chatbot for the NUML transport app, answering from a MySQL database, behind a FastAPI proxy with API-key authentication.

- Added in-app visualisations of chatbot responses, sentiment results and basic analytics.

WhatsApp Automation for Small Restaurants (n8n) *2025*

- Built an end-to-end WhatsApp ordering workflow in n8n, using the Gemini API as the LLM and Google Sheets as the datastore.
- Implemented an AI agent with conversation memory for order intake, confirmations and status updates.

Education

BS in Artificial Intelligence *2022 – 2026*
National University of Modern Languages (NUML), Islamabad

FSC (Pre-Engineering) *2019 – 2021*
Askaria College, Rawalpindi
Major: Physics, Chemistry, Mathematics

Certifications

- Buildables Fellowship (Deep Learning/Machine Learning Track) – Completion Certificate